

March 11, 2025
File No. 27225089.01

Compliance and Enforcement Section
Bureau of Air
Kansas Department of Health and Environment
1000 SW Jackson, Suite 310
Topeka, Kansas 66612

Subject: **Salina Municipal Solid Waste Landfill Facility**
2024 Annual NMOC Emission Rate Report
Source ID No. 1690059

To Whom It May Concern:

On behalf of the City of Salina, Kansas, and the Salina Municipal Solid Waste Landfill (Salina Landfill), SCS Engineers (SCS) is submitting the 2024 Annual Non-methane Organic Compound (NMOC) Emission Rate Report for the Salina Landfill. Because the Salina Landfill is subject to the Federal Implementation Plan 40 CFR 62.16724(c), the facility is required to calculate and submit its NMOC emission rate annually based on actual waste acceptance until an active gas collection system is required and installed to comply with 40 CFR 62 Subpart 000.

The annual NMOC emissions from the landfill are calculated using the LandGEM model version 3.02. The model combines site-specific NMOC concentrations from previous Tier 2 testing along with actual in-place waste data to calculate the total NMOC emissions for the year. The 2024 calculations used the following data:

- Actual 2024 waste acceptance: 84,491 megagrams (Mg)
- Total waste in place: 3,546,254 Mg
- Site-Specific NMOC concentration: 207 ppmv as Hexane (April 2021 Tier 2)

The resulting NMOC emission rate for the calendar year 2024 was calculated to be 18.32 Mg/year. A copy of the LandGEM model output is attached to this letter. This emission rate remains below the 34 Mg/year threshold for installing an active gas extraction system under Subpart 000.



March 11, 2025

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If you have any questions regarding this report or calculation, please do not hesitate to contact Marilyn Jones at 913-749-0715.

Sincerely,

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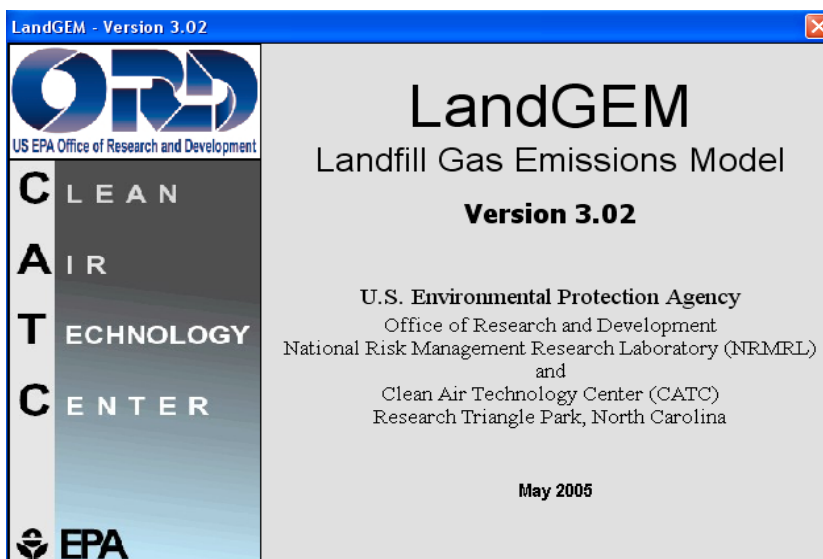
Allyson Patterson
Associate Professional
SCS Engineers

Handwritten signature of Marilyn Jones in blue ink.

Marilyn Jones
Senior Project Professional
SCS Engineers

cc: Tilden Crable, Salina MSWLF
Tiffany Luna, Salina MSWLF
Jim Kowach, Salina MSWLF

Attachment: LandGEM Model Output



Summary Report

Landfill Name or Identifier: Salina Municipal Solid Waste Landfill

Date: Monday, March 10, 2025

Description/Comments:

About LandGEM:

First-Order Decomposition Rate Equation:

$$Q_{CH_4} = \sum_{i=1}^n \sum_{j=0.1}^1 k L_o \left(\frac{M_i}{10} \right) e^{-k t_{ij}}$$

Where,

Q_{CH_4} = annual methane generation in the year of the calculation ($m^3/year$)

i = 1-year time increment

n = (year of the calculation) - (initial year of waste acceptance)

j = 0.1-year time increment

k = methane generation rate ($year^{-1}$)

L_o = potential methane generation capacity (m^3/Mg)

M_i = mass of waste accepted in the i^{th} year (Mg)

t_{ij} = age of the j^{th} section of waste mass M_i accepted in the i^{th} year (*decimal years*, e.g., 3.2 years)

LandGEM is based on a first-order decomposition rate equation for quantifying emissions from the decomposition of landfilled waste in municipal solid waste (MSW) landfills. The software provides a relatively simple approach to estimating landfill gas emissions. Model defaults are based on empirical data from U.S. landfills. Field test data can also be used in place of model defaults when available. Further guidance on EPA test methods, Clean Air Act (CAA) regulations, and other guidance regarding landfill gas emissions and control technology requirements can be found at <http://www.epa.gov/ttnatw01/landfill/landflpg.html>.

LandGEM is considered a screening tool — the better the input data, the better the estimates. Often, there are limitations with the available data regarding waste quantity and composition, variation in design and operating practices over time, and changes occurring over time that impact the emissions potential. Changes to landfill operation, such as operating under wet conditions through leachate recirculation or other liquid additions, will result in generating more gas at a faster rate. Defaults for estimating emissions for this type of operation are being developed to include in LandGEM along with defaults for conventional landfills (no leachate or liquid additions) for developing emission inventories and determining CAA applicability. Refer to the Web site identified above for future updates.

Input Review

LANDFILL CHARACTERISTICS

Landfill Open Year	1977	
Landfill Closure Year (with 80-year limit)	2056	Landfill Closure Year entered exceeds the 80-year waste acceptance limit. See Section 2.6 of the User's Manual.
Actual Closure Year (without limit)	2200	
Have Model Calculate Closure Year?	No	
Waste Design Capacity	16,013,644	megagrams

MODEL PARAMETERS

Methane Generation Rate, k	0.050	year ⁻¹
Potential Methane Generation Capacity, L ₀	170	m ³ /Mg
NMOC Concentration	207	ppmv as hexane
Methane Content	50	% by volume

GASES / POLLUTANTS SELECTED

Gas / Pollutant #1:	Total landfill gas
Gas / Pollutant #2:	Methane
Gas / Pollutant #3:	Carbon dioxide
Gas / Pollutant #4:	NMOC

WASTE ACCEPTANCE RATES

Year	Waste Accepted		Waste-In-Place	
	(Mg/year)	(short tons/year)	(Mg)	(short tons)
1977	37,160	40,876	0	0
1978	61,490	67,639	37,160	40,876
1979	67,250	73,975	98,650	108,515
1980	59,200	65,120	165,900	182,490
1981	53,600	58,960	225,100	247,610
1982	55,200	60,720	278,700	306,570
1983	56,700	62,370	333,900	367,290
1984	65,000	71,500	390,600	429,660
1985	63,400	69,740	455,600	501,160
1986	60,200	66,220	519,000	570,900
1987	72,200	79,420	579,200	637,120
1988	67,500	74,250	651,400	716,540
1989	64,100	70,510	718,900	790,790
1990	71,400	78,540	783,000	861,300
1991	77,600	85,360	854,400	939,840
1992	81,000	89,100	932,000	1,025,200
1993	75,000	82,500	1,013,000	1,114,300
1994	79,000	86,900	1,088,000	1,196,800
1995	82,000	90,200	1,167,000	1,283,700
1996	85,000	93,500	1,249,000	1,373,900
1997	77,000	84,700	1,334,000	1,467,400
1998	75,000	82,500	1,411,000	1,552,100
1999	73,000	80,300	1,486,000	1,634,600
2000	73,000	80,300	1,559,000	1,714,900
2001	69,000	75,900	1,632,000	1,795,200
2002	75,105	82,616	1,701,000	1,871,100
2003	80,188	88,207	1,776,105	1,953,716
2004	81,791	89,970	1,856,293	2,041,922
2005	79,904	87,894	1,938,084	2,131,892
2006	90,759	99,835	2,017,988	2,219,787
2007	91,287	100,416	2,108,747	2,319,622
2008	90,252	99,278	2,200,034	2,420,037
2009	87,126	95,839	2,290,286	2,519,315
2010	86,122	94,734	2,377,413	2,615,154
2011	87,276	96,003	2,463,534	2,709,888
2012	94,337	103,770	2,550,810	2,805,891
2013	95,287	104,815	2,645,147	2,909,661
2014	90,117	99,129	2,740,433	3,014,477
2015	75,165	82,682	2,830,551	3,113,606
2016	77,340	85,074	2,905,716	3,196,288

WASTE ACCEPTANCE RATES (Continued)

Year	Waste Accepted		Waste-In-Place	
	(Mg/year)	(short tons/year)	(Mg)	(short tons)
2017	83,473	91,820	2,983,056	3,281,362
2018	76,976	84,674	3,066,529	3,373,182
2019	77,641	85,405	3,143,505	3,457,856
2020	78,857	86,743	3,221,147	3,543,261
2021	79,508	87,458	3,300,004	3,630,004
2022	84,040	92,444	3,379,512	3,717,463
2023	82,702	90,973	3,463,551	3,809,906
2024	84,314	92,745	3,546,254	3,900,879
2025	86,843	95,528	3,630,568	3,993,624
2026	89,448	98,393	3,717,411	4,089,152
2027	92,132	101,345	3,806,859	4,187,545
2028	94,896	104,385	3,898,991	4,288,890
2029	97,743	107,517	3,993,887	4,393,276
2030	97,743	107,517	4,091,630	4,500,793
2031	97,743	107,517	4,189,373	4,608,310
2032	97,743	107,517	4,287,115	4,715,827
2033	97,743	107,517	4,384,858	4,823,344
2034	97,743	107,517	4,482,601	4,930,861
2035	97,743	107,517	4,580,344	5,038,378
2036	97,743	107,517	4,678,086	5,145,895
2037	97,743	107,517	4,775,829	5,253,412
2038	97,743	107,517	4,873,572	5,360,929
2039	97,743	107,517	4,971,315	5,468,446
2040	97,743	107,517	5,069,057	5,575,963
2041	97,743	107,517	5,166,800	5,683,480
2042	97,743	107,517	5,264,543	5,790,997
2043	97,743	107,517	5,362,286	5,898,514
2044	97,743	107,517	5,460,029	6,006,031
2045	97,743	107,517	5,557,771	6,113,548
2046	97,743	107,517	5,655,514	6,221,065
2047	97,743	107,517	5,753,257	6,328,583
2048	97,743	107,517	5,851,000	6,436,100
2049	97,743	107,517	5,948,742	6,543,617
2050	97,743	107,517	6,046,485	6,651,134
2051	97,743	107,517	6,144,228	6,758,651
2052	97,743	107,517	6,241,971	6,866,168
2053	97,743	107,517	6,339,713	6,973,685
2054	97,743	107,517	6,437,456	7,081,202
2055	97,743	107,517	6,535,199	7,188,719
2056	97,743	107,517	6,632,942	7,296,236

Pollutant Parameters

Gas / Pollutant Default Parameters:

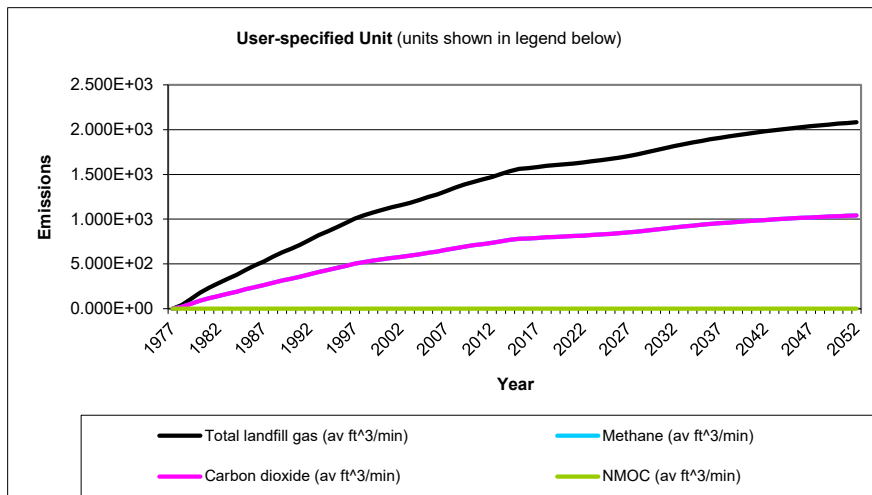
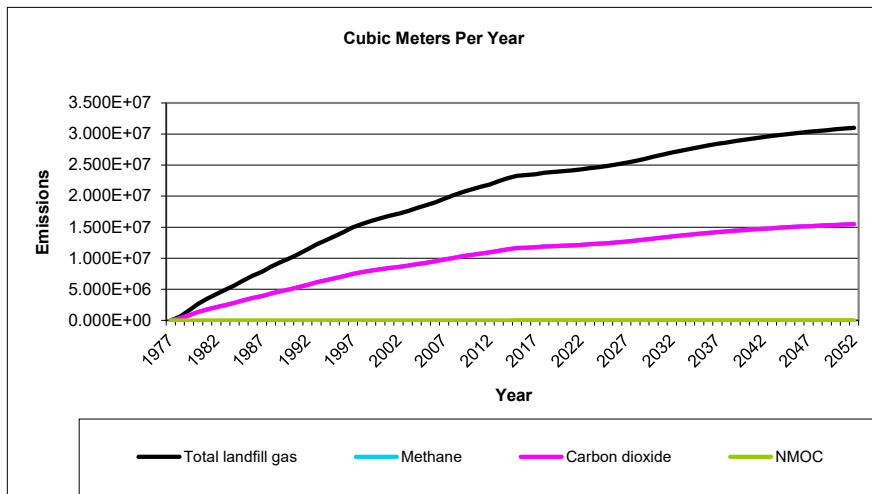
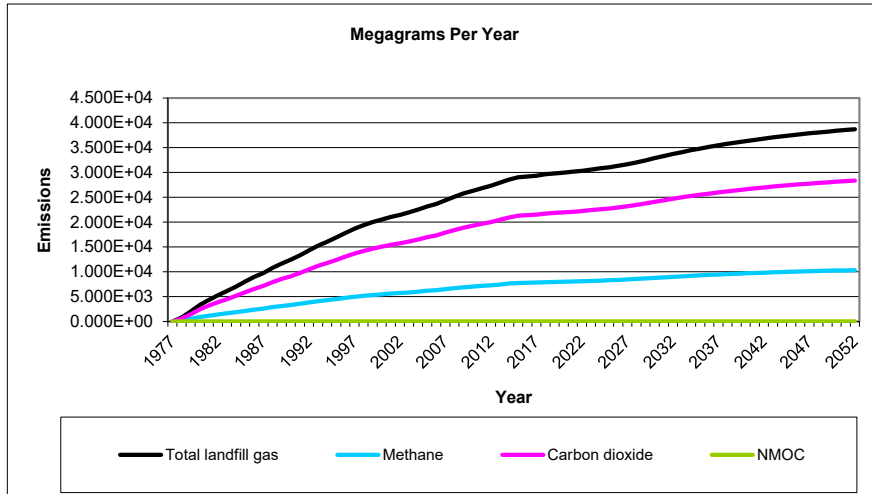
User-specified Pollutant Parameters:

	Compound	Concentration (ppmv)	Molecular Weight	Concentration (ppmv)	Molecular Weight
Gases	Total landfill gas		0.00		
	Methane		16.04		
	Carbon dioxide		44.01		
	NMOC		86.18		
Pollutants	1,1,1-Trichloroethane (methyl chloroform) - HAP	0.48	133.41		
	1,1,2,2- Tetrachloroethane - HAP/VOC	1.1	167.85		
	1,1-Dichloroethane (ethylidene dichloride) - HAP/VOC	2.4	98.97		
	1,1-Dichloroethene (vinylidene chloride) - HAP/VOC	0.20	96.94		
	1,2-Dichloroethane (ethylene dichloride) - HAP/VOC	0.41	98.96		
	1,2-Dichloropropane (propylene dichloride) - HAP/VOC	0.18	112.99		
	2-Propanol (isopropyl alcohol) - VOC	50	60.11		
	Acetone	7.0	58.08		
	Acrylonitrile - HAP/VOC	6.3	53.06		
	Benzene - No or Unknown Co-disposal - HAP/VOC	1.9	78.11		
	Benzene - Co-disposal - HAP/VOC	11	78.11		
	Bromodichloromethane - VOC	3.1	163.83		
	Butane - VOC	5.0	58.12		
	Carbon disulfide - HAP/VOC	0.58	76.13		
	Carbon monoxide	140	28.01		
	Carbon tetrachloride - HAP/VOC	4.0E-03	153.84		
	Carbonyl sulfide - HAP/VOC	0.49	60.07		
	Chlorobenzene - HAP/VOC	0.25	112.56		
	Chlorodifluoromethane	1.3	86.47		
	Chloroethane (ethyl chloride) - HAP/VOC	1.3	64.52		
	Chloroform - HAP/VOC	0.03	119.39		
	Chloromethane - VOC	1.2	50.49		
	Dichlorobenzene - (HAP for para isomer/VOC)	0.21	147		
	Dichlorodifluoromethane	16	120.91		
	Dichlorofluoromethane - VOC	2.6	102.92		
	Dichloromethane (methylene chloride) - HAP	14	84.94		
	Dimethyl sulfide (methyl sulfide) - VOC	7.8	62.13		
	Ethane	890	30.07		
	Ethanol - VOC	27	46.08		

Pollutant Parameters (Continued)

Gas / Pollutant Default Parameters:				User-specified Pollutant Parameters:	
	Compound	Concentration (ppmv)	Molecular Weight	Concentration (ppmv)	Molecular Weight
Pollutants	Ethyl mercaptan (ethanethiol) - VOC	2.3	62.13		
	Ethylbenzene - HAP/VOC	4.6	106.16		
	Ethylene dibromide - HAP/VOC	1.0E-03	187.88		
	Fluorotrichloromethane - VOC	0.76	137.38		
	Hexane - HAP/VOC	6.6	86.18		
	Hydrogen sulfide	36	34.08		
	Mercury (total) - HAP	2.9E-04	200.61		
	Methyl ethyl ketone - HAP/VOC	7.1	72.11		
	Methyl isobutyl ketone - HAP/VOC	1.9	100.16		
	Methyl mercaptan - VOC	2.5	48.11		
	Pentane - VOC	3.3	72.15		
	Perchloroethylene (tetrachloroethylene) - HAP	3.7	165.83		
	Propane - VOC	11	44.09		
	t-1,2-Dichloroethene - VOC	2.8	96.94		
	Toluene - No or Unknown Co-disposal - HAP/VOC	39	92.13		
	Toluene - Co-disposal - HAP/VOC	170	92.13		
	Trichloroethylene (trichloroethene) - HAP/VOC	2.8	131.40		
	Vinyl chloride - HAP/VOC	7.3	62.50		
	Xylenes - HAP/VOC	12	106.16		

Graphs



Results

Year	Total landfill gas			Methane		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
1977	0	0	0	0	0	0
1978	7.714E+02	6.177E+05	4.151E+01	2.061E+02	3.089E+05	2.075E+01
1979	2.010E+03	1.610E+06	1.082E+02	5.370E+02	8.049E+05	5.408E+01
1980	3.308E+03	2.649E+06	1.780E+02	8.837E+02	1.325E+06	8.900E+01
1981	4.376E+03	3.504E+06	2.354E+02	1.169E+03	1.752E+06	1.177E+02
1982	5.275E+03	4.224E+06	2.838E+02	1.409E+03	2.112E+06	1.419E+02
1983	6.164E+03	4.936E+06	3.316E+02	1.646E+03	2.468E+06	1.658E+02
1984	7.040E+03	5.638E+06	3.788E+02	1.881E+03	2.819E+06	1.894E+02
1985	8.046E+03	6.443E+06	4.329E+02	2.149E+03	3.222E+06	2.165E+02
1986	8.970E+03	7.183E+06	4.826E+02	2.396E+03	3.591E+06	2.413E+02
1987	9.782E+03	7.833E+06	5.263E+02	2.613E+03	3.917E+06	2.632E+02
1988	1.080E+04	8.652E+06	5.813E+02	2.886E+03	4.326E+06	2.906E+02
1989	1.168E+04	9.352E+06	6.283E+02	3.119E+03	4.676E+06	3.142E+02
1990	1.244E+04	9.961E+06	6.693E+02	3.323E+03	4.981E+06	3.346E+02
1991	1.332E+04	1.066E+07	7.164E+02	3.557E+03	5.331E+06	3.582E+02
1992	1.428E+04	1.143E+07	7.681E+02	3.813E+03	5.716E+06	3.841E+02
1993	1.526E+04	1.222E+07	8.211E+02	4.077E+03	6.111E+06	4.106E+02
1994	1.607E+04	1.287E+07	8.649E+02	4.294E+03	6.436E+06	4.324E+02
1995	1.693E+04	1.356E+07	9.109E+02	4.522E+03	6.779E+06	4.555E+02
1996	1.781E+04	1.426E+07	9.581E+02	4.757E+03	7.130E+06	4.790E+02
1997	1.870E+04	1.498E+07	1.006E+03	4.996E+03	7.488E+06	5.031E+02
1998	1.939E+04	1.553E+07	1.043E+03	5.179E+03	7.763E+06	5.216E+02
1999	2.000E+04	1.602E+07	1.076E+03	5.343E+03	8.008E+06	5.381E+02
2000	2.054E+04	1.645E+07	1.105E+03	5.487E+03	8.224E+06	5.526E+02
2001	2.105E+04	1.686E+07	1.133E+03	5.624E+03	8.430E+06	5.664E+02
2002	2.146E+04	1.718E+07	1.155E+03	5.732E+03	8.592E+06	5.773E+02
2003	2.197E+04	1.759E+07	1.182E+03	5.869E+03	8.797E+06	5.911E+02
2004	2.257E+04	1.807E+07	1.214E+03	6.028E+03	9.035E+06	6.071E+02
2005	2.316E+04	1.855E+07	1.246E+03	6.187E+03	9.274E+06	6.231E+02
2006	2.369E+04	1.897E+07	1.275E+03	6.329E+03	9.486E+06	6.374E+02
2007	2.442E+04	1.956E+07	1.314E+03	6.523E+03	9.778E+06	6.570E+02
2008	2.513E+04	2.012E+07	1.352E+03	6.711E+03	1.006E+07	6.759E+02
2009	2.577E+04	2.064E+07	1.387E+03	6.884E+03	1.032E+07	6.933E+02
2010	2.633E+04	2.108E+07	1.416E+03	7.032E+03	1.054E+07	7.082E+02
2011	2.683E+04	2.148E+07	1.443E+03	7.166E+03	1.074E+07	7.217E+02
2012	2.733E+04	2.189E+07	1.471E+03	7.301E+03	1.094E+07	7.353E+02
2013	2.796E+04	2.239E+07	1.504E+03	7.468E+03	1.119E+07	7.521E+02
2014	2.857E+04	2.288E+07	1.537E+03	7.632E+03	1.144E+07	7.686E+02
2015	2.905E+04	2.326E+07	1.563E+03	7.760E+03	1.163E+07	7.815E+02
2016	2.919E+04	2.338E+07	1.571E+03	7.798E+03	1.169E+07	7.853E+02
2017	2.938E+04	2.352E+07	1.580E+03	7.846E+03	1.176E+07	7.902E+02
2018	2.968E+04	2.376E+07	1.597E+03	7.927E+03	1.188E+07	7.983E+02
2019	2.983E+04	2.388E+07	1.605E+03	7.967E+03	1.194E+07	8.024E+02
2020	2.998E+04	2.401E+07	1.613E+03	8.009E+03	1.200E+07	8.066E+02
2021	3.016E+04	2.415E+07	1.623E+03	8.056E+03	1.207E+07	8.113E+02
2022	3.034E+04	2.429E+07	1.632E+03	8.104E+03	1.215E+07	8.161E+02
2023	3.060E+04	2.451E+07	1.647E+03	8.174E+03	1.225E+07	8.233E+02
2024	3.083E+04	2.469E+07	1.659E+03	8.234E+03	1.234E+07	8.293E+02
2025	3.107E+04	2.488E+07	1.672E+03	8.300E+03	1.244E+07	8.359E+02
2026	3.136E+04	2.511E+07	1.687E+03	8.377E+03	1.256E+07	8.437E+02

Results (Continued)

Year	Total landfill gas			Methane		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2027	3.169E+04	2.538E+07	1.705E+03	8.464E+03	1.269E+07	8.525E+02
2028	3.206E+04	2.567E+07	1.725E+03	8.563E+03	1.283E+07	8.623E+02
2029	3.246E+04	2.599E+07	1.747E+03	8.671E+03	1.300E+07	8.733E+02
2030	3.291E+04	2.635E+07	1.771E+03	8.790E+03	1.318E+07	8.853E+02
2031	3.333E+04	2.669E+07	1.793E+03	8.904E+03	1.335E+07	8.967E+02
2032	3.374E+04	2.701E+07	1.815E+03	9.011E+03	1.351E+07	9.075E+02
2033	3.412E+04	2.732E+07	1.836E+03	9.114E+03	1.366E+07	9.179E+02
2034	3.449E+04	2.761E+07	1.855E+03	9.211E+03	1.381E+07	9.277E+02
2035	3.483E+04	2.789E+07	1.874E+03	9.304E+03	1.395E+07	9.370E+02
2036	3.516E+04	2.816E+07	1.892E+03	9.392E+03	1.408E+07	9.459E+02
2037	3.548E+04	2.841E+07	1.909E+03	9.476E+03	1.420E+07	9.544E+02
2038	3.578E+04	2.865E+07	1.925E+03	9.556E+03	1.432E+07	9.624E+02
2039	3.606E+04	2.888E+07	1.940E+03	9.632E+03	1.444E+07	9.701E+02
2040	3.633E+04	2.909E+07	1.955E+03	9.704E+03	1.455E+07	9.773E+02
2041	3.659E+04	2.930E+07	1.969E+03	9.773E+03	1.465E+07	9.843E+02
2042	3.683E+04	2.949E+07	1.982E+03	9.838E+03	1.475E+07	9.908E+02
2043	3.707E+04	2.968E+07	1.994E+03	9.901E+03	1.484E+07	9.971E+02
2044	3.729E+04	2.986E+07	2.006E+03	9.960E+03	1.493E+07	1.003E+03
2045	3.750E+04	3.003E+07	2.017E+03	1.002E+04	1.501E+07	1.009E+03
2046	3.770E+04	3.019E+07	2.028E+03	1.007E+04	1.509E+07	1.014E+03
2047	3.789E+04	3.034E+07	2.038E+03	1.012E+04	1.517E+07	1.019E+03
2048	3.807E+04	3.048E+07	2.048E+03	1.017E+04	1.524E+07	1.024E+03
2049	3.824E+04	3.062E+07	2.058E+03	1.021E+04	1.531E+07	1.029E+03
2050	3.841E+04	3.075E+07	2.066E+03	1.026E+04	1.538E+07	1.033E+03
2051	3.856E+04	3.088E+07	2.075E+03	1.030E+04	1.544E+07	1.037E+03
2052	3.871E+04	3.100E+07	2.083E+03	1.034E+04	1.550E+07	1.041E+03
2053	3.885E+04	3.111E+07	2.090E+03	1.038E+04	1.556E+07	1.045E+03
2054	3.899E+04	3.122E+07	2.098E+03	1.041E+04	1.561E+07	1.049E+03
2055	3.911E+04	3.132E+07	2.104E+03	1.045E+04	1.566E+07	1.052E+03
2056	3.924E+04	3.142E+07	2.111E+03	1.048E+04	1.571E+07	1.055E+03
2057	3.935E+04	3.151E+07	2.117E+03	1.051E+04	1.576E+07	1.059E+03
2058	3.743E+04	2.997E+07	2.014E+03	9.998E+03	1.499E+07	1.007E+03
2059	3.561E+04	2.851E+07	1.916E+03	9.511E+03	1.426E+07	9.578E+02
2060	3.387E+04	2.712E+07	1.822E+03	9.047E+03	1.356E+07	9.111E+02
2061	3.222E+04	2.580E+07	1.733E+03	8.606E+03	1.290E+07	8.667E+02
2062	3.065E+04	2.454E+07	1.649E+03	8.186E+03	1.227E+07	8.244E+02
2063	2.915E+04	2.334E+07	1.568E+03	7.787E+03	1.167E+07	7.842E+02
2064	2.773E+04	2.220E+07	1.492E+03	7.407E+03	1.110E+07	7.460E+02
2065	2.638E+04	2.112E+07	1.419E+03	7.046E+03	1.056E+07	7.096E+02
2066	2.509E+04	2.009E+07	1.350E+03	6.702E+03	1.005E+07	6.750E+02
2067	2.387E+04	1.911E+07	1.284E+03	6.375E+03	9.556E+06	6.421E+02
2068	2.270E+04	1.818E+07	1.222E+03	6.064E+03	9.090E+06	6.108E+02
2069	2.160E+04	1.729E+07	1.162E+03	5.769E+03	8.647E+06	5.810E+02
2070	2.054E+04	1.645E+07	1.105E+03	5.487E+03	8.225E+06	5.526E+02
2071	1.954E+04	1.565E+07	1.051E+03	5.220E+03	7.824E+06	5.257E+02
2072	1.859E+04	1.488E+07	1.000E+03	4.965E+03	7.442E+06	5.000E+02
2073	1.768E+04	1.416E+07	9.513E+02	4.723E+03	7.079E+06	4.757E+02
2074	1.682E+04	1.347E+07	9.049E+02	4.493E+03	6.734E+06	4.525E+02
2075	1.600E+04	1.281E+07	8.608E+02	4.273E+03	6.406E+06	4.304E+02
2076	1.522E+04	1.219E+07	8.188E+02	4.065E+03	6.093E+06	4.094E+02
2077	1.448E+04	1.159E+07	7.789E+02	3.867E+03	5.796E+06	3.894E+02

Results (Continued)

Year	Total landfill gas			Methane		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2078	1.377E+04	1.103E+07	7.409E+02	3.678E+03	5.513E+06	3.704E+02
2079	1.310E+04	1.049E+07	7.047E+02	3.499E+03	5.244E+06	3.524E+02
2080	1.246E+04	9.977E+06	6.704E+02	3.328E+03	4.989E+06	3.352E+02
2081	1.185E+04	9.491E+06	6.377E+02	3.166E+03	4.745E+06	3.188E+02
2082	1.127E+04	9.028E+06	6.066E+02	3.011E+03	4.514E+06	3.033E+02
2083	1.072E+04	8.588E+06	5.770E+02	2.865E+03	4.294E+06	2.885E+02
2084	1.020E+04	8.169E+06	5.489E+02	2.725E+03	4.084E+06	2.744E+02
2085	9.704E+03	7.770E+06	5.221E+02	2.592E+03	3.885E+06	2.610E+02
2086	9.231E+03	7.391E+06	4.966E+02	2.466E+03	3.696E+06	2.483E+02
2087	8.780E+03	7.031E+06	4.724E+02	2.345E+03	3.515E+06	2.362E+02
2088	8.352E+03	6.688E+06	4.494E+02	2.231E+03	3.344E+06	2.247E+02
2089	7.945E+03	6.362E+06	4.275E+02	2.122E+03	3.181E+06	2.137E+02
2090	7.557E+03	6.052E+06	4.066E+02	2.019E+03	3.026E+06	2.033E+02
2091	7.189E+03	5.756E+06	3.868E+02	1.920E+03	2.878E+06	1.934E+02
2092	6.838E+03	5.476E+06	3.679E+02	1.827E+03	2.738E+06	1.840E+02
2093	6.505E+03	5.209E+06	3.500E+02	1.737E+03	2.604E+06	1.750E+02
2094	6.187E+03	4.955E+06	3.329E+02	1.653E+03	2.477E+06	1.664E+02
2095	5.886E+03	4.713E+06	3.167E+02	1.572E+03	2.356E+06	1.583E+02
2096	5.599E+03	4.483E+06	3.012E+02	1.495E+03	2.242E+06	1.506E+02
2097	5.326E+03	4.264E+06	2.865E+02	1.423E+03	2.132E+06	1.433E+02
2098	5.066E+03	4.056E+06	2.726E+02	1.353E+03	2.028E+06	1.363E+02
2099	4.819E+03	3.859E+06	2.593E+02	1.287E+03	1.929E+06	1.296E+02
2100	4.584E+03	3.670E+06	2.466E+02	1.224E+03	1.835E+06	1.233E+02
2101	4.360E+03	3.491E+06	2.346E+02	1.165E+03	1.746E+06	1.173E+02
2102	4.148E+03	3.321E+06	2.231E+02	1.108E+03	1.661E+06	1.116E+02
2103	3.945E+03	3.159E+06	2.123E+02	1.054E+03	1.580E+06	1.061E+02
2104	3.753E+03	3.005E+06	2.019E+02	1.002E+03	1.503E+06	1.010E+02
2105	3.570E+03	2.859E+06	1.921E+02	9.535E+02	1.429E+06	9.603E+01
2106	3.396E+03	2.719E+06	1.827E+02	9.070E+02	1.360E+06	9.135E+01
2107	3.230E+03	2.587E+06	1.738E+02	8.628E+02	1.293E+06	8.689E+01
2108	3.073E+03	2.460E+06	1.653E+02	8.207E+02	1.230E+06	8.266E+01
2109	2.923E+03	2.340E+06	1.573E+02	7.807E+02	1.170E+06	7.863E+01
2110	2.780E+03	2.226E+06	1.496E+02	7.426E+02	1.113E+06	7.479E+01
2111	2.645E+03	2.118E+06	1.423E+02	7.064E+02	1.059E+06	7.114E+01
2112	2.516E+03	2.014E+06	1.353E+02	6.719E+02	1.007E+06	6.767E+01
2113	2.393E+03	1.916E+06	1.287E+02	6.392E+02	9.581E+05	6.437E+01
2114	2.276E+03	1.823E+06	1.225E+02	6.080E+02	9.113E+05	6.123E+01
2115	2.165E+03	1.734E+06	1.165E+02	5.784E+02	8.669E+05	5.825E+01
2116	2.060E+03	1.649E+06	1.108E+02	5.501E+02	8.246E+05	5.541E+01
2117	1.959E+03	1.569E+06	1.054E+02	5.233E+02	7.844E+05	5.270E+01

Results (Continued)

Year	Carbon dioxide			NMOC		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
1977	0	0	0	0	0	0
1978	5.654E+02	3.089E+05	2.075E+01	4.583E-01	1.279E+02	8.592E-03
1979	1.473E+03	8.049E+05	5.408E+01	1.194E+00	3.332E+02	2.239E-02
1980	2.425E+03	1.325E+06	8.900E+01	1.966E+00	5.484E+02	3.685E-02
1981	3.207E+03	1.752E+06	1.177E+02	2.600E+00	7.254E+02	4.874E-02
1982	3.866E+03	2.112E+06	1.419E+02	3.134E+00	8.744E+02	5.875E-02
1983	4.518E+03	2.468E+06	1.658E+02	3.662E+00	1.022E+03	6.865E-02
1984	5.160E+03	2.819E+06	1.894E+02	4.183E+00	1.167E+03	7.841E-02
1985	5.897E+03	3.222E+06	2.165E+02	4.781E+00	1.334E+03	8.961E-02
1986	6.574E+03	3.591E+06	2.413E+02	5.330E+00	1.487E+03	9.990E-02
1987	7.169E+03	3.917E+06	2.632E+02	5.812E+00	1.622E+03	1.089E-01
1988	7.918E+03	4.326E+06	2.906E+02	6.419E+00	1.791E+03	1.203E-01
1989	8.559E+03	4.676E+06	3.142E+02	6.939E+00	1.936E+03	1.301E-01
1990	9.117E+03	4.981E+06	3.346E+02	7.391E+00	2.062E+03	1.385E-01
1991	9.759E+03	5.331E+06	3.582E+02	7.911E+00	2.207E+03	1.483E-01
1992	1.046E+04	5.716E+06	3.841E+02	8.483E+00	2.366E+03	1.590E-01
1993	1.119E+04	6.111E+06	4.106E+02	9.068E+00	2.530E+03	1.700E-01
1994	1.178E+04	6.436E+06	4.324E+02	9.551E+00	2.664E+03	1.790E-01
1995	1.241E+04	6.779E+06	4.555E+02	1.006E+01	2.806E+03	1.886E-01
1996	1.305E+04	7.130E+06	4.790E+02	1.058E+01	2.952E+03	1.983E-01
1997	1.371E+04	7.488E+06	5.031E+02	1.111E+01	3.100E+03	2.083E-01
1998	1.421E+04	7.763E+06	5.216E+02	1.152E+01	3.214E+03	2.159E-01
1999	1.466E+04	8.008E+06	5.381E+02	1.188E+01	3.315E+03	2.228E-01
2000	1.505E+04	8.224E+06	5.526E+02	1.220E+01	3.405E+03	2.288E-01
2001	1.543E+04	8.430E+06	5.664E+02	1.251E+01	3.490E+03	2.345E-01
2002	1.573E+04	8.592E+06	5.773E+02	1.275E+01	3.557E+03	2.390E-01
2003	1.610E+04	8.797E+06	5.911E+02	1.306E+01	3.642E+03	2.447E-01
2004	1.654E+04	9.035E+06	6.071E+02	1.341E+01	3.740E+03	2.513E-01
2005	1.698E+04	9.274E+06	6.231E+02	1.376E+01	3.839E+03	2.580E-01
2006	1.736E+04	9.486E+06	6.374E+02	1.408E+01	3.927E+03	2.639E-01
2007	1.790E+04	9.778E+06	6.570E+02	1.451E+01	4.048E+03	2.720E-01
2008	1.841E+04	1.006E+07	6.759E+02	1.493E+01	4.165E+03	2.798E-01
2009	1.889E+04	1.032E+07	6.933E+02	1.531E+01	4.272E+03	2.870E-01
2010	1.929E+04	1.054E+07	7.082E+02	1.564E+01	4.364E+03	2.932E-01
2011	1.966E+04	1.074E+07	7.217E+02	1.594E+01	4.447E+03	2.988E-01
2012	2.003E+04	1.094E+07	7.353E+02	1.624E+01	4.531E+03	3.044E-01
2013	2.049E+04	1.119E+07	7.521E+02	1.661E+01	4.634E+03	3.114E-01
2014	2.094E+04	1.144E+07	7.686E+02	1.698E+01	4.736E+03	3.182E-01
2015	2.129E+04	1.163E+07	7.815E+02	1.726E+01	4.815E+03	3.235E-01
2016	2.140E+04	1.169E+07	7.853E+02	1.735E+01	4.839E+03	3.251E-01
2017	2.153E+04	1.176E+07	7.902E+02	1.745E+01	4.869E+03	3.272E-01
2018	2.175E+04	1.188E+07	7.983E+02	1.763E+01	4.919E+03	3.305E-01
2019	2.186E+04	1.194E+07	8.024E+02	1.772E+01	4.944E+03	3.322E-01
2020	2.197E+04	1.200E+07	8.066E+02	1.781E+01	4.970E+03	3.339E-01
2021	2.210E+04	1.207E+07	8.113E+02	1.792E+01	4.999E+03	3.359E-01
2022	2.223E+04	1.215E+07	8.161E+02	1.803E+01	5.029E+03	3.379E-01
2023	2.243E+04	1.225E+07	8.233E+02	1.818E+01	5.073E+03	3.408E-01
2024	2.259E+04	1.234E+07	8.293E+02	1.832E+01	5.110E+03	3.433E-01
2025	2.277E+04	1.244E+07	8.359E+02	1.846E+01	5.151E+03	3.461E-01
2026	2.298E+04	1.256E+07	8.437E+02	1.863E+01	5.198E+03	3.493E-01

Results (Continued)

Year	Carbon dioxide			NMOC		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2027	2.322E+04	1.269E+07	8.525E+02	1.883E+01	5.253E+03	3.529E-01
2028	2.349E+04	1.283E+07	8.623E+02	1.905E+01	5.313E+03	3.570E-01
2029	2.379E+04	1.300E+07	8.733E+02	1.929E+01	5.381E+03	3.615E-01
2030	2.412E+04	1.318E+07	8.853E+02	1.955E+01	5.455E+03	3.665E-01
2031	2.443E+04	1.335E+07	8.967E+02	1.980E+01	5.525E+03	3.712E-01
2032	2.472E+04	1.351E+07	9.075E+02	2.004E+01	5.592E+03	3.757E-01
2033	2.501E+04	1.366E+07	9.179E+02	2.027E+01	5.656E+03	3.800E-01
2034	2.527E+04	1.381E+07	9.277E+02	2.049E+01	5.716E+03	3.841E-01
2035	2.553E+04	1.395E+07	9.370E+02	2.070E+01	5.774E+03	3.879E-01
2036	2.577E+04	1.408E+07	9.459E+02	2.089E+01	5.828E+03	3.916E-01
2037	2.600E+04	1.420E+07	9.544E+02	2.108E+01	5.881E+03	3.951E-01
2038	2.622E+04	1.432E+07	9.624E+02	2.126E+01	5.930E+03	3.984E-01
2039	2.643E+04	1.444E+07	9.701E+02	2.142E+01	5.977E+03	4.016E-01
2040	2.663E+04	1.455E+07	9.773E+02	2.159E+01	6.022E+03	4.046E-01
2041	2.681E+04	1.465E+07	9.843E+02	2.174E+01	6.065E+03	4.075E-01
2042	2.699E+04	1.475E+07	9.908E+02	2.188E+01	6.105E+03	4.102E-01
2043	2.716E+04	1.484E+07	9.971E+02	2.202E+01	6.144E+03	4.128E-01
2044	2.733E+04	1.493E+07	1.003E+03	2.215E+01	6.180E+03	4.153E-01
2045	2.748E+04	1.501E+07	1.009E+03	2.228E+01	6.215E+03	4.176E-01
2046	2.763E+04	1.509E+07	1.014E+03	2.240E+01	6.249E+03	4.198E-01
2047	2.777E+04	1.517E+07	1.019E+03	2.251E+01	6.280E+03	4.220E-01
2048	2.790E+04	1.524E+07	1.024E+03	2.262E+01	6.310E+03	4.240E-01
2049	2.803E+04	1.531E+07	1.029E+03	2.272E+01	6.339E+03	4.259E-01
2050	2.815E+04	1.538E+07	1.033E+03	2.282E+01	6.366E+03	4.277E-01
2051	2.826E+04	1.544E+07	1.037E+03	2.291E+01	6.392E+03	4.295E-01
2052	2.837E+04	1.550E+07	1.041E+03	2.300E+01	6.416E+03	4.311E-01
2053	2.847E+04	1.556E+07	1.045E+03	2.308E+01	6.440E+03	4.327E-01
2054	2.857E+04	1.561E+07	1.049E+03	2.316E+01	6.462E+03	4.342E-01
2055	2.867E+04	1.566E+07	1.052E+03	2.324E+01	6.483E+03	4.356E-01
2056	2.876E+04	1.571E+07	1.055E+03	2.331E+01	6.503E+03	4.370E-01
2057	2.884E+04	1.576E+07	1.059E+03	2.338E+01	6.523E+03	4.383E-01
2058	2.743E+04	1.499E+07	1.007E+03	2.224E+01	6.205E+03	4.169E-01
2059	2.610E+04	1.426E+07	9.578E+02	2.116E+01	5.902E+03	3.965E-01
2060	2.482E+04	1.356E+07	9.111E+02	2.012E+01	5.614E+03	3.772E-01
2061	2.361E+04	1.290E+07	8.667E+02	1.914E+01	5.340E+03	3.588E-01
2062	2.246E+04	1.227E+07	8.244E+02	1.821E+01	5.080E+03	3.413E-01
2063	2.137E+04	1.167E+07	7.842E+02	1.732E+01	4.832E+03	3.247E-01
2064	2.032E+04	1.110E+07	7.460E+02	1.648E+01	4.596E+03	3.088E-01
2065	1.933E+04	1.056E+07	7.096E+02	1.567E+01	4.372E+03	2.938E-01
2066	1.839E+04	1.005E+07	6.750E+02	1.491E+01	4.159E+03	2.794E-01
2067	1.749E+04	9.556E+06	6.421E+02	1.418E+01	3.956E+03	2.658E-01
2068	1.664E+04	9.090E+06	6.108E+02	1.349E+01	3.763E+03	2.529E-01
2069	1.583E+04	8.647E+06	5.810E+02	1.283E+01	3.580E+03	2.405E-01
2070	1.506E+04	8.225E+06	5.526E+02	1.221E+01	3.405E+03	2.288E-01
2071	1.432E+04	7.824E+06	5.257E+02	1.161E+01	3.239E+03	2.176E-01
2072	1.362E+04	7.442E+06	5.000E+02	1.104E+01	3.081E+03	2.070E-01
2073	1.296E+04	7.079E+06	4.757E+02	1.051E+01	2.931E+03	1.969E-01
2074	1.233E+04	6.734E+06	4.525E+02	9.993E+00	2.788E+03	1.873E-01
2075	1.173E+04	6.406E+06	4.304E+02	9.506E+00	2.652E+03	1.782E-01
2076	1.115E+04	6.093E+06	4.094E+02	9.042E+00	2.523E+03	1.695E-01
2077	1.061E+04	5.796E+06	3.894E+02	8.601E+00	2.400E+03	1.612E-01

Results (Continued)

Year	Carbon dioxide			NMOC		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2078	1.009E+04	5.513E+06	3.704E+02	8.182E+00	2.283E+03	1.534E-01
2079	9.600E+03	5.244E+06	3.524E+02	7.783E+00	2.171E+03	1.459E-01
2080	9.132E+03	4.989E+06	3.352E+02	7.403E+00	2.065E+03	1.388E-01
2081	8.686E+03	4.745E+06	3.188E+02	7.042E+00	1.965E+03	1.320E-01
2082	8.263E+03	4.514E+06	3.033E+02	6.699E+00	1.869E+03	1.256E-01
2083	7.860E+03	4.294E+06	2.885E+02	6.372E+00	1.778E+03	1.194E-01
2084	7.476E+03	4.084E+06	2.744E+02	6.061E+00	1.691E+03	1.136E-01
2085	7.112E+03	3.885E+06	2.610E+02	5.765E+00	1.608E+03	1.081E-01
2086	6.765E+03	3.696E+06	2.483E+02	5.484E+00	1.530E+03	1.028E-01
2087	6.435E+03	3.515E+06	2.362E+02	5.217E+00	1.455E+03	9.779E-02
2088	6.121E+03	3.344E+06	2.247E+02	4.962E+00	1.384E+03	9.302E-02
2089	5.823E+03	3.181E+06	2.137E+02	4.720E+00	1.317E+03	8.848E-02
2090	5.539E+03	3.026E+06	2.033E+02	4.490E+00	1.253E+03	8.417E-02
2091	5.269E+03	2.878E+06	1.934E+02	4.271E+00	1.192E+03	8.006E-02
2092	5.012E+03	2.738E+06	1.840E+02	4.063E+00	1.133E+03	7.616E-02
2093	4.767E+03	2.604E+06	1.750E+02	3.865E+00	1.078E+03	7.244E-02
2094	4.535E+03	2.477E+06	1.664E+02	3.676E+00	1.026E+03	6.891E-02
2095	4.314E+03	2.356E+06	1.583E+02	3.497E+00	9.756E+02	6.555E-02
2096	4.103E+03	2.242E+06	1.506E+02	3.326E+00	9.280E+02	6.235E-02
2097	3.903E+03	2.132E+06	1.433E+02	3.164E+00	8.827E+02	5.931E-02
2098	3.713E+03	2.028E+06	1.363E+02	3.010E+00	8.397E+02	5.642E-02
2099	3.532E+03	1.929E+06	1.296E+02	2.863E+00	7.987E+02	5.367E-02
2100	3.359E+03	1.835E+06	1.233E+02	2.723E+00	7.598E+02	5.105E-02
2101	3.196E+03	1.746E+06	1.173E+02	2.591E+00	7.227E+02	4.856E-02
2102	3.040E+03	1.661E+06	1.116E+02	2.464E+00	6.875E+02	4.619E-02
2103	2.891E+03	1.580E+06	1.061E+02	2.344E+00	6.540E+02	4.394E-02
2104	2.750E+03	1.503E+06	1.010E+02	2.230E+00	6.221E+02	4.180E-02
2105	2.616E+03	1.429E+06	9.603E+01	2.121E+00	5.917E+02	3.976E-02
2106	2.489E+03	1.360E+06	9.135E+01	2.018E+00	5.629E+02	3.782E-02
2107	2.367E+03	1.293E+06	8.689E+01	1.919E+00	5.354E+02	3.597E-02
2108	2.252E+03	1.230E+06	8.266E+01	1.826E+00	5.093E+02	3.422E-02
2109	2.142E+03	1.170E+06	7.863E+01	1.737E+00	4.845E+02	3.255E-02
2110	2.038E+03	1.113E+06	7.479E+01	1.652E+00	4.608E+02	3.096E-02
2111	1.938E+03	1.059E+06	7.114E+01	1.571E+00	4.384E+02	2.945E-02
2112	1.844E+03	1.007E+06	6.767E+01	1.495E+00	4.170E+02	2.802E-02
2113	1.754E+03	9.581E+05	6.437E+01	1.422E+00	3.966E+02	2.665E-02
2114	1.668E+03	9.113E+05	6.123E+01	1.352E+00	3.773E+02	2.535E-02
2115	1.587E+03	8.669E+05	5.825E+01	1.286E+00	3.589E+02	2.411E-02
2116	1.509E+03	8.246E+05	5.541E+01	1.224E+00	3.414E+02	2.294E-02
2117	1.436E+03	7.844E+05	5.270E+01	1.164E+00	3.247E+02	2.182E-02